

MIAE 215: Computer Programming for Engineers

Arduino Hardware Kit Specifications & Laboratory Academic Policies

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering

Instructor: Prof. Brandon W. Gordon · Language: Embedded C++ (Arduino Core)

1. Official Course Laboratory Regulations

The laboratory component of **MIAE 215** provides practical experience in physical computing, digital input/output, analog sensor signal acquisition, and servo actuation on the Atmel ATmega328P platform.

MIAE 215 LABORATORY TIMELINE & MILESTONES		
Period	Laboratory Module & Scope	Deliverable Type
Weeks 1-3	Hardware Kit Acquisition & Setup	Independent Prep
Week 4	Lab 1: Software Setup & Blink Lifecycle	Optional/Practice
Week 6	Lab 2: Digital I/O & Speed Profiling	Optional/Practice
Week 8	Lab 3: Analog ADC & Servo Kinematics	Optional/Practice
Week 9+	Arduino Mechatronics Term Project	MANDATORY MOODLE
Final	Final Exam Embedded C++ Questions	10% OF COURSE GR.

Key Academic Policies

- 1. Individual Work Mandate:** Every student must complete the labs and the term project independently. Collaboration or code copying between students is strictly forbidden and subject to academic integrity review.
- 2. Attendance Policy:**
 - Laboratory attendance in scheduled campus rooms is **completely optional**.
 - All lab manuals, video demonstrations, and official solution sets are published online.
 - If you are able to perform the experiments on your personal computer and hardware kit, you are not required to attend in person.
 - In-person sessions function as open drop-in support hours with Teaching Assistants (TAs).
- 3. Section Flexibility:** Students who need in-person assistance may attend alternative lab sections provided physical seating and bench equipment are available.
- 4. Graded Deliverables:**
 - **Labs 1, 2, and 3 are NOT submitted for grades.** They are self-study practice modules with full solutions provided.
 - **Only the Arduino Term Project is submitted for a course grade** via Moodle.
- 5. Final Exam Impact (10%):**
 - **Approximately 10% of the MIAE 215 Final Examination** consists of circuit analysis and embedded C++ coding questions drawn directly from the Arduino labs and project concepts.

2. Recommended Hardware Starter Kits

The Arduino kit works with both **Windows** and **macOS** operating systems. Prof. Gordon recommends acquiring one of the following starter kits:

Kit Description	Amazon Canada Link	Included Essential Components
ELEGOO UNO Project Super Starter Kit	Amazon.ca B06XXYVWVJ	Arduino Uno R3, SG90 Servo, Photoresistors, LEDs, Resistors, Breadboard, Wires
SunFounder Starter Kit for Arduino	Amazon.ca B08B4JY95V	Uno R3, Ultrasonic Sensor, SG90 Servo, LDR Light Sensors, Buzzer, Breadboard
Rexqualis Complete Starter Kit	Amazon.ca B07BTB3N3J	Uno R3, Dual-axis Joystick, Stepper Motor, Servo, Resistor packs, Wires
Kuman Complete Starter Kit	Amazon.ca B01D8KQZF4	Uno R3, Multiple Sensors, Breadboard, Jumper cables, Power Supply Module

[!TIP] **Minimalist Component List:** If you already have Arduino hardware, you do not need to buy a new kit. For Labs 1–3, you need:

- 1x Arduino Uno (or compatible clone) with USB cable.
- 1x Solderless breadboard and male-to-male jumper wires.
- 1x SG90 9g micro-servo motor.
- 1x Photoresistor (LDR) and 10 k Ω pull-down resistor.
- 2x LEDs and 220 Ω current-limiting resistors.
- 1x Additional sensor of your choice for the Term Project (e.g., analog joystick, HC-SR04 ultrasonic sensor, or TMP36 temperature sensor).

3. Official Lab Download Resources

All official files provided by Prof. Gordon are hosted at:

https://users.encs.concordia.ca/~bwgordon/MIAE_215_labs.html

- **Lab 1 Files:** Google Drive archive ([Arduino_lab_introduction.rar](#)) containing video screencasts and starter code.
- **Lab 2 Files:** [arduino_lab_2.pdf](#) , [arduino_lab_2.rar](#) (starter code), and [arduino_lab_2_sol.rar](#) (solutions).
- **Lab 3 Files:** [arduino_lab_3.pdf](#) and [arduino_lab_3_sol.rar](#) (solutions).
- **Term Project:** [MIAE_215_arduino_project.rar](#) containing the project prompt document ([MIAE_215_arduino_project.doc](#)).